Antiderivatives of trigonometric fun[redacted]s

1

a $f(x) = -7\cos x + C$

b $f(x) = -7\sin x + C$

c $f(x) = -6\cos x - \sin x + C$

d $f(x) = -15\cos\left(\frac{x}{3}\right) - 8\sin\left(\frac{x}{4}\right) + C$

2

a $-\frac{1}{7}\cos 7x + C$

b $\frac{1}{6}\sin 6x + C$

c $-3\cos\left(\frac{x}{3}\right) + C$

d $-3\cos 3x + C$

e $-20\sin\left(\frac{x}{4}\right) + C$

f $\sin\left(x + \frac{\pi}{6}\right) + C$

g $-\cos\left(x - \frac{\pi}{6}\right) + C$

h $-\frac{1}{5}\cos\left(5x - \frac{\pi}{4}\right) + C$

i $8\sin\left(\frac{x}{8} + \frac{\pi}{3}\right) + C$

j $2\sin\left(4x - \frac{\pi}{3}\right) + C$

k $4\cos\left(2x + \frac{\pi}{6}\right) + C$

l $-14\cos\left(\frac{x}{7} - \frac{\pi}{4}\right) + C$

m $-28\sin\left(\frac{x}{7} + \frac{\pi}{4}\right) + C$

n $\frac{4x^{\frac{5}{4}}}{5} - \frac{1}{3}\cos 3x + C$

o $\sin x + \frac{1}{3}\cos 3x + C$

p $-3\cos\left(\frac{x}{3}\right) + 3\sin\left(\frac{x}{3}\right) + C$

q $-\cos 7x + 21\sin 3x + C$

3

a $\sin x + x\cos x$

b $4x\sin x + 4\cos x + C$

Equations of functions

4

a $y = 6\cos\left(\frac{x}{2}\right) + 11$

b $y = -\frac{1}{2}\cos 2x + \frac{1}{3}\sin 3x + \frac{23}{6}$

5

a $k = 2$

b $f(x) = \frac{2}{3}\sin 3x + \frac{16}{3}$

6

a $k = \frac{13}{4}$

b $f(x) = -13\cos\left(\frac{x}{4}\right) + 1$

7

a $4\sin 2x + C$

b $y = 4\sin 2x - 1$